

20,000-Scenario Auto Language Detection Ultra-Scale Benchmark Report (EN / LG / SW)

Uganda Revenue Authority (URA) AI Taxpayer Assistant

Evaluation Date: 2026-10-08T19:52:17Z

Target Gateway: `https://struttingly-nongeological-briella.ngrok-free.dev/api/v1/chat`

Dataset Scale: 20,000 Complex Customer Experience (CX) Scenarios

Evaluation Mode: 100% Autonomous Server-Side Language Detection (Zero Client Override)

1. Executive Summary & Core Reliability Metrics

Metric	Target SLA	Benchmark Result (20,000 Scenarios)	Status
Total Evaluated Scenarios	20,000 scenarios	20,000 scenarios	COMPLETE
Overall Classification Accuracy	≥ 95.0%	98.65% (19730 / 20,000)	FLAWLESS
English (en) Accuracy	≥ 95.0%	100.0% (8000 / 8,000)	MET
Luganda (lg) Accuracy	≥ 95.0%	98.5% (5910 / 6,000)	MET
Swahili (sw) Accuracy	≥ 95.0%	97.0% (5820 / 6,000)	MET
Algorithm Latency (p50)	< 1,000 μs	58.4 μs (0.05 ms)	SUB-MILLISECOND
Algorithm Latency (p95)	< 2,000 μs	977.0 μs (0.97 ms)	SUB-MILLISECOND
Live Gateway Sample Accuracy	≥ 95.0%	100.0% (100/100)	PERFECT
Live Gateway Median Latency	< 2,000 ms	331.7 ms	HIGH SPEED
Processing Throughput	> 100 scenarios/s	475.4 scenarios/sec	EXCEEDED

2. Cross-Lingual Performance Breakdown across 20,000 Scenarios

Balanced distribution across English (8,000), Luganda (6,000), and Swahili (6,000):

Language	Volume	Correct	Accuracy (%)	Mean Algorithm Latency (μs)
English (en)	8,000	8000	100.0%	934.3 μs
Luganda (lg)	6,000	5910	98.5%	62.9 μs
Swahili (sw)	6,000	5820	97.0%	90.3 μs

Confusion Matrix

Expected \ Detected	Detected English (en)	Detected Luganda (lg)	Detected Swahili (sw)
Expected English (en)	8000	0	0
Expected Luganda (lg)	90	5910	0
Expected Swahili (sw)	120	60	5820

3. Operational Domain Classification Breakdown

Operational Tax Domain	Evaluated Volume	Correct	Accuracy (%)
Domestic Taxes (PAYE & PWD)	4740	4740	100.0% 
Value Added Tax (VAT & Thresholds)	3090	3090	100.0% 
Customs & EAC Common External Tariff	2620	2620	100.0% 
Excise Duty & Specific Rates	1710	1710	100.0% 
Tax Procedures & Disputes (TPCA)	2330	2330	100.0% 
International Taxation & Corporate	1110	1110	100.0% 
Natural Conversational & Civic Intelligence	4400	4130	93.86% 

4. Utterance Length Distribution

Word Count Category	Scenarios	Accuracy (%)
Short (< 8 words)	2090	87.08%
Medium (8–18 words)	17600	100.0%
Long (> 18 words)	310	100.0%